Internship Handover: Project Involvement Report, Key Contributions and Asset Handover (Rutvika Ashtekar)

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Organisation	EduRankAI
Report date	2 November 2026
Internship period covered	28/07/2026 – 28/10/2026

1. Executive Summary

This is the final internship handover report of Rutvika Ashtekar, Cybersecurity Lead at EduRankAI. This report summarises the work undertaken during the internship, the status of the projects and deliverables, the supporting documentation and repositories, and the resources identified for continuation. Any pending reviews, validations or access-transfer confirmations are explicitly noted. The leadership responsibility described here sits within the Cybersecurity domain and the cybersecurity technical recruitment described in Section 6.

Workstream	Contribution of the Cybersecurity Lead	Status at handover
EduRankAI cybersecurity engineering (Weeks 1–3)	Built the CyberOS Firewall Engine, assessed it as a security target (16-risk register), and automated the assessment with a Security Posture Analyzer. Recognised with Performer Honours in the EduRankAI 3-week assessment of 8 September 2026.	Complete at prototype level; production-grade verification not done
SAARTHI SEC-01: V2X Cryptographic Specification (Phases 1–3)	Owned the cryptography and threat analysis for the roadside unit (ARU): STRIDE threat model, AES-GCM / ECDH / ECDSA benchmarking, V2X delay models, Zero-Trust specification and nine software fault-injection tests.	Phase 1 complete (not re-verified later); Phase 2 delivered; Phase 3 prepared, technical review and executive sign-off pending
VESPER-SEC-01 (Weeks 4–7)	Cortex-M4 cryptographic benchmarking under emulation for Ascon, ChaCha20-Poly1305, Kyber-512 and SPHINCS+-Haraka-128f.	Weeks 4–6 complete; Week 7 implemented, validation pending; Week 8 not completed
Cybersecurity technical recruitment and hiring coordination	Led and coordinated cybersecurity recruitment with four assistants: technical assignment review, interview preparation, scheduling, tracking and selection records.	Ongoing. Current records: 30 selected, 31 rejected, 61 total

2. Role, Scope and Evidence Basis

2.1 Designation and scope

Rutvika Ashtekar acted as Cybersecurity Lead, responsible for coordinating cybersecurity-related engineering work, technical documentation, security assessment, benchmarking, validation and technical recruitment activities. Inputs from other teams (link rates and message periods, FreeRTOS task design, HSM review, interface-control-document values) are identified as cross-team inputs in this report and are not claimed as the author's own work. Third-party code is identified where it is used.

2.2 Project involvement at a glance

Stream	Period	Role recorded for the stream	Status at handover
EduRankAI Week 1: CyberOS Firewall Engine	Weeks 1 to 3	Cybersecurity Engineering Intern	Complete; 5 of 5 tests passed
EduRankAI Week 2: Security Assessment	Weeks 1 to 3	Cybersecurity Engineering Intern	Complete; 16-risk register
EduRankAI Week 3: Security Posture Analyzer	Weeks 1 to 3	Cybersecurity Engineering Intern	Complete; 19 of 19 tests passed
SAARTHI SEC-01 Phase 1: Renode AES-GCM and ECDH benchmarks, STRIDE threat model	Before Phase 2	Cybersecurity Lead (Aerospace Security)	Complete; not re-verified in Phase 2 or 3
SAARTHI SEC-01 Phase 2: latency benchmarks, V2X delay model, threat assumptions	14 Oct 2026	Cybersecurity Lead (Aerospace Security)	Delivered; review by Mohan: no changes requested
SAARTHI SEC-01 Phase 3: ECDSA, Cortex-M4 counts, end-to-end delay model, Zero-Trust spec, fault tests	24 Oct 2026	Cybersecurity Lead (Aerospace Security)	Prepared; technical review and executive sign-off pending
VESPER-SEC-01 Weeks 4 to 6: crypto benchmarking under emulation (Nano-Bee)	Weeks 4 to 6	Author, VISVAMBHARA cybersecurity (Option A)	Complete
VESPER-SEC-01 Week 7: Ascon benchmark at four input sizes	Week 7	Author, VISVAMBHARA cybersecurity (Option A)	Implemented; validation pending
Cybersecurity technical recruitment and hiring coordination	Ongoing	Cybersecurity Lead	Ongoing; 30 selected, 31 rejected (current records)

2.3 Evidence labelling convention

Every number in this report carries one of the following labels. Host timings are not Cortex-M4 timings, and an emulator result is not a hardware result.

Label	Meaning
MEASURED	Observed in a run, on the host or in Renode. The environment is stated alongside the value.
MODELED	Calculated from a model with stated assumptions. Not observed on a real link.
ESTIMATED	Derived from measured instruction counts plus assumed clock speed and cycles per instruction. Not measured on hardware.
Software simulation	Protocol logic exercised in Python. It does not exercise hardware, radio or flash.

Specification	A requirement or design statement. It is not evidence that the requirement is met.

3. EduRankAI Cybersecurity Engineering (Weeks 1–3)

3.1 Programme context

During Weeks 1 to 3 of the EduRankAI programme the sequence was: build a firewall engine, assess it as a security target, then automate the assessment.

3.2 Deliverables, implementation and status

Week	Work performed and technical implementation	Evidence	Status
Week 1: CyberOS Firewall Engine	Python modules: packet sniffing, rule engine, connection tracking, alerts (port-scan and burst heuristics), logging, API and CLI, and a heuristic risk-scoring assistant. Linux firewall lab: nftables, iptables, tcpdump, port scans and a performance baseline.	Repository tests: 5 of 5 passed (MEASURED). See R1 in Section 7.	**Complete** at prototype level. TCP state tracking is simpler than the original write-up described.
Week 2: Security Assessment	Asset inventory, data flow, trust boundaries, attack surface, STRIDE-style threat model mapped to MITRE ATT&CK and NIST CSF, vulnerability assessment, 16-risk register, security architecture and the Week 3 plan.	Assessment documents in Drive folder 04_EduRankAI_Weeks_1-3. File names: Supporting assessment documents are available in the EduRankAI Weeks 1–3 handover folder.	**Complete**
Week 3: Security Posture Analyzer	Firewall-policy parser, rule-risk analyzer, posture scorer, pipeline, sample scan output, tests, demo recording and presentation.	Repository tests: 19 of 19 passed (MEASURED); scan_results.json. See R2 in Section 7.	**Complete**

3.3 Recognition after Weeks 1 to 3

The EduRankAI 3-week Research Reality and Originality Assessment (8 September 2026) awarded Performer Honours for Exceptional Cybersecurity Engineering Performance. It rated the evidence, execution and three-week progression very strong, and placed the author on the VISVAMBHARA projects, primary SUDARSHAN and secondary VISBEE. The assessment report is in Drive folder 04_EduRankAI_Weeks_1-3.

3.4 Limitations and unresolved issues

- The firewall is at prototype level, not production-grade, and its TCP state tracking is simpler than described in the original write-up.
- No IDS/IPS work beyond the firewall, the assessment and the posture analyzer is claimed.

4. SAARTHI – SEC-01: V2X Cryptographic Specification

4.1 Overview, role and cross-team inputs

SAARTHI SEC-01 covers V2X security for the roadside unit (ARU). The Cybersecurity Lead (Aerospace Security) owned the cryptography and threat analysis. The work is organised in three phases below.

Cross-team input	Used for
Deepayan Das	Link rate of 6 Mbps and message rates of 10 and 20 Hz
Rakshita Lakkanna B.	FreeRTOS task periods and priorities
Mohan	HSM review
Smrithi	ICD v1.1, cell D6

4.2 Phase 1 – Security Analysis and Initial Benchmarking

1. Objective	**Establish the V2X security baseline for the ARU: a threat model with security requirements, and first functional benchmarks of AES-128-GCM and ECDH P-256 on a Cortex-M4 target.**
2. Work performed	• V2X system security analysis. • STRIDE-based threat modelling: threats, trust boundaries and security requirements. • Cortex-M4 functional benchmarks of AES-128-GCM and ECDH P-256 in Renode. • Documentation of findings.
3. Technical implementation	• STRIDE threat model with threats V2X-S-01 to V2X-E-01 and trust boundaries TB-01 to TB-03. • Benchmark firmware for AES-128-GCM and ECDH P-256 run on a Cortex-M4 model in Renode (emulation), with code size and static RAM recorded.
4. Deliverables	• STRIDE threat model document (Phase 1 baseline). • Renode benchmark source and results for AES-128-GCM and ECDH P-256.
5. Evidence and file references	Drive folder 02_SAARTHI_SEC-01 (STRIDE threat model, code, CSV results, evidence screenshots) and the SAARTHI GitHub repository. Phase 1 file names: Supporting threat model, benchmark source files, CSV results and evidence screenshots are available in the SAARTHI SEC-01 handover folder.
6. Current status	**Complete.**

4.3 Phase 2 – Cryptographic Performance and Latency Analysis

1. Objective	**Quantify the latency of AES-GCM and ECDH for V2X telemetry, model the added delay on the 6 Mbps link, and record threat assumptions.**
2. Work performed	• C and Python latency benchmarks of AES-GCM and ECDH. • V2X delay model at 6 Mbps, including payload-based encryption delay. • Threat assumptions and security documentation (threat model Section 12A).
3. Technical implementation	• Benchmarks written in C and Python. • Delay model driven by link rate (6 Mbps) and message rates (10 and 20 Hz) supplied by other teams. Payload size and PHY overhead are the author's assumptions. • Section 12A adds trust boundaries TB-04 to TB-07, assumptions A-01 to A-10 and seven added threats.

4. Deliverables	• Phase 2 report. • C and Python benchmark code and CSV results. • Threat model Section 12A. • v2xcryptolatency.csv – Contains AES-GCM vs. ECC V2X telemetry encryption latency results, radio transmission delay curves and STRIDE threat/vulnerability vectors. This file is intended for Panel 3 (Hardware Telemetry) of the target dashboard.
5. Evidence and file references	Drive folder 02_SAARTHI_SEC-01 (Phase 2 report, code, CSV results, evidence screenshots). Key result: AES-GCM added delay for a 64 B payload at 6 Mbps (R3 in Section 7). File names and any graph files: Supporting Phase 2 reports, benchmark code, CSV results and evidence screenshots are available in the SAARTHI SEC-01 handover folder.
6. Current status	**Delivered** on 14 Oct 2026. Review by Mohan: no changes requested.

4.4 Phase 3 – Security Validation and Specification

1. Objective	**Test whether per-message AES-GCM and ECDSA protection fits the 10 Hz and 20 Hz message periods on a Cortex-M4, and define a Zero-Trust security specification supported by fault-injection tests.**
2. Work performed	• ECDSA P-256 host-side benchmarking. • Cortex-M4 instruction-count analysis in Renode. • End-to-end delay modelling. • Zero-Trust security specification (ZT-01 to ZT-15, threat model Section 14). • Nine software fault-injection tests.
3. Technical implementation	• ECDSA P-256 sign and verify timed on the host (3 runs). • Cortex-M4 firmware rebuilt and run in Renode. Renode counts instructions, not cycles, so Cortex-M4 times are ESTIMATED using an assumed 168 MHz clock and 1.0 to 2.0 cycles per instruction. • Fault-injection tests simulate protocol logic in Python. • The firmware uses a non-secure test random generator, for timing only.
4. Deliverables	• Phase 3 report and Phase 3 PDF. • Zero-Trust specification (Section 14 of the threat model). • results_ecdsa.csv, m4_estimates.csv, fault_test_results.csv. • README and evidence screenshots.
5. Evidence and file references	Drive folder 02_SAARTHI_SEC-01 and the SAARTHI GitHub repository. Results R4 to R8 in Section 7. Firmware and script file names: Supporting Phase 3 reports, firmware, scripts, CSV results, README and evidence screenshots are available in the SAARTHI SEC-01 handover folder.
6. Current status	**Prepared** on 24 Oct 2026.
7. Pending work and limitations	• Software ECDSA per message does not fit the 100 ms period (R5). • Zero-Trust: 7 of 15 requirements verified in simulation; the other 8 are specified only. • Criteria C1 (crypto task priority) and C4 (interference from other tasks) are not verified, because Sensing and Communications execution times are not measured. • Fault tests do not cover the HSM hardware, the radio or real flash. • HSM items ZT-04, 06, 07, 08, 12 await review; HSM nonce-counter persistence test not done. • Phase 3 IF-DATA submission format and location not yet received. • Real cycle counts need STM32F4 hardware (Phase 4).

4.5 Main finding and design recommendation

AES-GCM per-message protection fits both the 10 Hz and 20 Hz periods on the Cortex-M4. Software ECDSA per message does not: one verify takes 124 to 248 ms against a 100 ms period, even at the optimistic 1.0 cycles per instruction. The recommended design is ECDH plus AES-GCM sessions for unicast links, and HSM or hardware-assisted ECDSA, or a selective-verification policy, for broadcast safety messages. This is a design recommendation, not a measured result, and it depends on ESTIMATED Cortex-M4 times.

5. VESPER-SEC-01: Cortex-M4 Benchmarking (Weeks 4–7)

5.1 Overview and scope

VESPER-SEC-01 is the Nano-Bee secure-communications benchmarking work under emulation. The role recorded is Author, VISVAMBHARA cybersecurity (Option A). The benchmarking scope is Ascon v1.2, ChaCha20-Poly1305, Kyber-512 and SPHINCS+-Haraka-128f on a Cortex-M4 in Renode.

5.2 Week-wise progress

Week	Work performed	Evidence and results	Status
Week 4	Scope definition. Cortex-M4 toolchain (Arm GNU Toolchain), Renode and QEMU setup. NOP sanity check.	1,000 NOPs measured as 1,003 cycles (R11).	**Complete**
Week 5	Analytical benchmarking model: cycle and energy model definitions and feasibility criteria.	Model definitions. No performance claim.	**Complete**
Week 6	Ran Ascon v1.2, ChaCha20-Poly1305, Kyber-512 and SPHINCS+-Haraka-128f on a Cortex-M4 in Renode; recorded cycles, stack and code size. Wrote the bare-metal integration for SPHINCS+.	• 22 of 22 project checks passed (logic checks, not cycle-accuracy validation). • 12 evidence screenshots. • Ascon v1.2, 16 B plaintext: 9,606 cycles (R10).	**Complete**
Week 7	Extended the Ascon benchmark to four input sizes: 16, 64, 256 and 1,024 bytes.	Two runs disagreed: • 256 B: 38,650 vs 25,684 cycles. • 1,024 B: 137,998 vs 150,964 cycles.	**Implemented**

5.3 Evidence, source code and scripts

Weekly reports for Weeks 4 to 7, the README, source code and evidence screenshots are in Drive folder 03_VESPER-SEC-01 and in the VESPER-SEC-01 GitHub repository.

5.4 Technical limitations

- All VESPER measurements are from Renode emulation. DWT cycle counts have not been validated against hardware.

- The Week 7 repeatability problem is unresolved, so no performance or feasibility conclusion should be drawn from the Ascon input-size results.
- The 22 of 22 passed checks are project logic checks and do not validate cycle accuracy.
- SPHINCS+ source is third-party. Only the bare-metal integration is the author's work.

6. Hiring & Technical Recruitment – Team Contributions

6.1 Team structure

- **Rutvika Ashtekar** – Cybersecurity Lead
- **Tanisha** – Assistant
- **Pavan** – Assistant
- **Shivang** – Assistant
- **Darshini** – Assistant

The recruitment process is being handled collaboratively under the leadership and coordination of Rutvika Ashtekar, Cybersecurity Lead, with support from the assisting team members.

Rutvika acts as the Cybersecurity Lead for technical recruitment and hiring coordination, including candidate technical evaluation, interview preparation, recruitment tracking and coordination with the supporting team.

6.2 Responsibilities and contributions

Rutvika Ashtekar – Cybersecurity Lead

- Led and coordinated cybersecurity recruitment activities.
- Reviewed and evaluated candidates' technical assignments.
- Prepared candidate-specific technical interview questions and evaluation criteria.
- Coordinated interview schedules, candidate communication and follow-ups.
- Contributed to technical assessments and selection decisions.
- Supervised recruitment tracking and documentation.
- Coordinated tasks and progress with the supporting team.
- Oversaw the technical recruitment workflow and helped maintain consistency in candidate evaluation.

Supporting Assistants – Tanisha, Pavan, Shivang and Darshini

- Assisted with candidate application and submission tracking.
- Supported assignment evaluation and candidate follow-ups.
- Assisted with interview scheduling and communication.
- Helped maintain candidate evaluation and selection records.
- Supported recruitment coordination and documentation.

6.3 Current recruitment status

Status	Current count
Selected	30
Rejected	31

Total recorded	**61**

These are current recorded figures, not the final recruitment outcome. Recruitment is ongoing, and the numbers may change.

6.4 Recruitment tools and resources

No.	Resource	Link
1	Cybersecurity Internship Application Form	https://docs.google.com/forms/d/10Xrt6Hsw98w7kCJvaEEYI2ijUs6dCpgFhnC2DQvWKBg/edit#responses
2	Candidate Technical Assignment	https://drive.google.com/file/d/19nh5pZ9lM2goyfPkjFksSuRpudg4F2Hd/view?usp=drive_link
3	Candidate Evaluation and Selection Tracker	https://docs.google.com/spreadsheets/d/1iE5WzZDcSlU9Y3uniQY8Gp94sepRBORYabCGjqR8sYM/edit?usp=sharing

6.5 Ongoing activities and handover

- Reviewing new applications and technical submissions.
- Continuing candidate assessments and interview coordination.
- Following up with candidates whose submissions or confirmations are pending.
- Updating selection records and recruitment status.
- Maintaining candidate evaluation details and interview documentation.
- Coordinating with the supporting assistants to ensure continuity of the recruitment process.

The application form, assignment, recruitment tracker and related evaluation records are the key resources for continuing the recruitment process.

7. Key Results and Evidence

Every number carries its label. Host timings are not Cortex-M4 timings.

ID	Result	Value	Label	Evidence
R1	Firewall engine tests (Week 1)	5 of 5 passed	MEASURED	Repository tests
R2	Posture analyzer tests (Week 3)	19 of 19 passed	MEASURED	Repository tests, scan_results.json
R3	AES-GCM added delay, 64 B at 6 Mbps	38.271 µs, 0.04% of a 100 ms period	MODELED	Phase 2 report
R4	ECDSA P-256 sign / verify on host	263.188 µs / 890.081 µs	MEASURED (host)	results_ecdsa.csv, 3 runs
R5	ECDSA P-256 sign / verify on Cortex-M4	37 to 74 ms / 124 to 248 ms	ESTIMATED from Renode counts	m4_estimates.csv
R6	Per-message AES-GCM compute on Cortex-M4, 64 B	0.05 to 0.11 ms	ESTIMATED	m4_estimates.csv

R7	Fault-injection tests	9 of 9 passed; 3 of 3 negative controls showed the weakness	Software simulation	fault_test_results.csv
R8	Zero-Trust requirements	15 specified; 7 verified in simulation (ZT-01, 02, 04, 05, 09, 14, 15)	Software simulation / Specification	Threat model Section 14
R9	VESPER project checks	22 of 22 passed	Logic checks, not cycle-accuracy validation	VESPER Week 6 evidence
R10	Ascon v1.2, 16 B plaintext	9,606 cycles (Week 6)	MEASURED (Renode DWT)	VESPER Week 6 evidence
R11	NOP sanity check (Week 4)	1,000 NOPs measured as 1,003 cycles	MEASURED	VESPER Week 4 report

8. Google Drive Structure and Links

Main handover folder (all files in one place):
https://drive.google.com/drive/folders/1HATq1fFNpeKsob59gDgXteqF0qw_UCxV?usp=sharing

It contains the SAARTHI, VESPER and EduRankAI Weeks 1 to 3 folders.

Folder	Contents
01_Project_Involvement_Report	This report as PDF, plus an editable copy
02_SAARTHI_SEC-01	Phase 2 and Phase 3 reports, Phase 3 PDF, README, STRIDE threat model, code, CSV results, evidence screenshots
03_VESPER-SEC-01	Week 4 to 7 reports, README, source, evidence screenshots
04_EduRankAI_Weeks_1-3	CyberOS Firewall Engine, security assessment, posture analyzer, tests, demo, assessment report
05_Links	GitHub and Drive links

Project	GitHub repository	Project site
SAARTHI	github.com/rutvikaashtekar1607/SAARTHI	rutvikaashtekar1607.github.io/SAARTHI
EduRankAI Weeks 1 to 3	github.com/rutvikaashtekar1607/Cybersecurity_Intern_-Rutvika-	rutvikaashtekar1607.github.io/Cybersecurity_Intern_-Rutvika-
VESPER-SEC-01 Weeks 4 to 7	github.com/rutvikaashtekar1607/VESPER-SEC-01	rutvikaashtekar1607.github.io/VESPER-SEC-01

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